

UK Biobank Patient Record

$\{\text{Demographics \& Lifestyle}\}$
age · sex · BMI
smoking · alcohol
living alone

$\{\text{Blood Biomarkers} \times 21\}$
HbA1c · cholesterol
HDL/LDL · CRP
creatinine ...

$\{\text{Comorbidity Flags} \times 11\}$
T2 diabetes · CKD
depression · stroke
hypertension ...

$\{\text{Temporal ICD History}\}$
J45 asthma @ 44yr
F32 depression @ 52yr
E11 diabetes @ 58yr

Delphi
(zero-shot LLM)

Baseline
CoxPH

Disease Text
Emb + CoxPH

Trajectory
Emb + CoxPH

Clinical
Summary Emb

Input format

Discrete ICD token
sequence (age-binned)
[D₁ D₂ ... D_n □?]

39 numeric features
[age, sex, BMI,
21 labs, 11 flags]

Disease-only prose
"At age 44, patient
diagnosed with J45..."

Temporal token
stream (ICD + age)
"44yr:J45 → 52yr:F32..."

Full clinical prose
demographics + labs
+ disease history

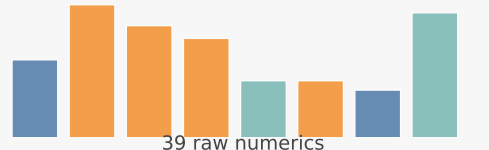
Encoder/
architecture

Autoregressive LLM (Delphi)



causal attention · death-token logit
→ sigmoid hazard chain → S(t)

No encoder (direct CoxPH fit)



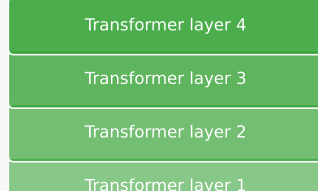
39 raw numerics
directly enter Cox model

Qwen3-8B (encoder mode)



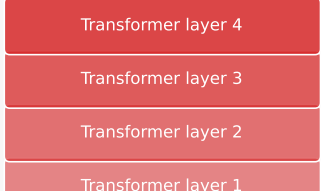
bidirectional · mean-pool
→ 4096-dim sentence vector

Qwen3-8B (encoder mode)



bidirectional · mean-pool
→ 4096-dim sentence vector

Qwen3-8B (encoder mode)



bidirectional · mean-pool
→ 4096-dim sentence vector

Features &
survival
model

Survival curve S(t)
from hazard chain
(no CoxPH)

39-dim vector
→ CoxPH
(L2 penalizer)

PCA(128) + 39 raw
= 167-dim
→ CoxPH

PCA(128) + 39 raw
= 167-dim
→ CoxPH

PCA(128) only
= 128-dim
→ CoxPH
(no raw features)

Output

S(t) survival curve
(zero-shot)

Risk score
ranking

Risk score
ranking

Risk score
ranking

Risk score
ranking

Data modality colour key:



Demographics & lifestyle



Blood biomarkers



Comorbidity flags



Temporal ICD history



All modalities (fused into text)